COAST: Altitude as a Coupled Perception Variable in Shared-Camera Aerial-Ground Tracking

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Abstract: In shared-camera aerial-ground tracking, a single downward-facing UAV camera is the ground vehicle's only exteroceptive sensor, serving at once as its navigation reference through lane segmentation and as its state estimator through marker detection and an extended Kalman filter. We observe that this shared sensor couples navigation and estimation through one scalar, the UAV altitude, which sets localization resolution (ground-sampling noise $R \propto z^2$) and navigation coverage (camera footprint) in opposite directions. Treating altitude as an active perception variable, we characterize the coupling in closed form: the estimation-optimal altitude is a curvature-scheduled coverage floor, and adapting altitude strictly improves estimation over any fixed altitude if and only if that floor binds and the path curvature varies. We deploy the resulting policy, COAST (Coupled-Altitude Shared-camera Tracking), with a belief-space ground controller and a perception-triggered aerial planner whose ancillary law is input-to-state stable between triggers and free of Zeno behavior. A reproducible ROS 2/Gazebo study verifies each prediction: the measured sensor noise follows $R \propto z^2$, the steady-state estimation cost is monotone and track-independent in altitude, the curvature-scheduled policy matches the best feasible fixed altitude in the non-binding regime of our testbed, and the perception-driven event trigger reduces aerial-replanning computation by more than an order of magnitude at matched estimation quality. The code and data are publicly available.

Keywords: Perception-aware control, Cooperative aerial-ground systems, Belief-space planning, Event-triggered control, Estimation and Kalman filtering

1. INTRODUCTION

Unmanned ground vehicles (UGVs) operating in unstructured or GPS-denied environments are limited by short and occlusion-prone perception horizons: onboard LiDAR and stereo cameras have a narrow field of view and cannot see far enough ahead to anticipate the upcoming path (Frey et al., 2024). A now-standard remedy is aerial-ground cooperation, in which an unmanned aerial vehicle (UAV) acts as an elevated eye for the perception-limited ground robot (Grocholsky et al., 2006; Li et al., 2024; Santos et al., 2024). Most such systems, however, hold the UAV at a fixed altitude and never treat that altitude as a decision variable: the aerial vehicle follows a distance-minimizing trajectory and returns the ground robot's state estimate as a downstream byproduct. This decoupling is fragile. When the ground robot maneuvers sharply or is briefly occluded, the estimation covariance grows and tracking degrades, precisely when reliable sensing matters most.

We begin instead from a structural observation about the shared sensor. When a single downward-facing camera on the UAV is the ground robot's only exteroceptive sensor, one image serves two roles at once: the lane ahead is segmented into a navigation waypoint, and a marker on the robot is detected into a noisy pose measurement for the estimator. Because both products are read from the same image, they share the same imaging geometry, and in particular the same altitude. Raising the camera enlarges the ground footprint and improves navigation coverage, but it coarsens the ground sampling distance and inflates the localization noise as $R \propto z^2$; lowering it does the reverse. A single scalar, the UAV altitude, therefore sets estimation resolution and navigation coverage simultaneously and in opposite directions. This antagonism turns altitude from a geometric parameter into an active perception variable, and characterizing how to use it is the subject of this paper.

This viewpoint connects to two established lines of work, and departs from both in the same place. Perception-aware trajectory optimization shapes a robot's own motion to serve estimation, through the sensor trajectory (Falanga et al., 2018; Spasojevic et al., 2020) and viewpoint selection (Zhang and Scaramuzza, 2019), while belief-space planning and active SLAM go further, augmenting the planning state with the estimator covariance to act deliberately under partial observability (Platt Jr. et al., 2010; van den Berg et al., 2012; Placed et al., 2023). Our ground-robot controller belongs to this belief-space family. What is new is where the information decision lives and what it trades: in these works a robot reshapes its own trajectory to see better, whereas here the decisive variable is a teammate's altitude, and it sets two perception

[†] denotes equal contribution.

products against each other rather than sharpening one. That altitude alone trades localization resolution against sensor coverage is itself known, and has been exploited to balance a coverage-quality criterion in multi-UAV area coverage (Papatheodorou et al., 2017) and to meet resolution constraints in photogrammetric coverage planning (Di Franco and Buttazzo, 2016). What has not, to our knowledge, been formulated is the closed-loop *shared-camera* instantiation, where one downward camera is the ground robot's only exteroceptive sensor and the same altitude sets both an EKF measurement (favoring low z) and lane-segmentation coverage (favoring higher z). For this coupling we give a closed-form, curvature-scheduled optimal altitude and characterize when adapting it beats any fixed altitude.

Quadrotor target tracking under field-of-view and occlusion limits is likewise well studied (Thomas et al., 2017; Penin et al., 2018), but there too the coverage-versus-accuracy trade is split across separate robots or sensors and the altitude is held fixed rather than optimized.

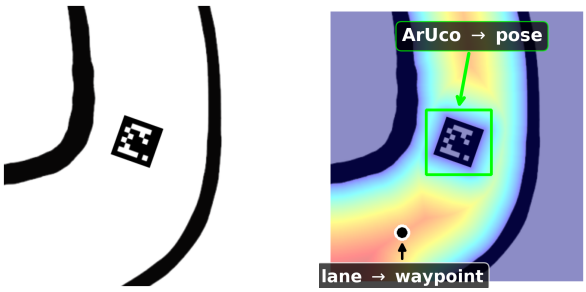
Deploying the resulting policy in closed loop additionally demands robust tracking under bounded disturbances and a bounded compute budget. Tube MPC supplies the first, decomposing control into a nominal plan and an ancillary law that confines the error to a robust positively invariant tube (Mayne et al., 2005; Raković et al., 2005), and event-triggered control supplies the second, re-solving only when a state-dependent condition fires (Tabuada, 2007; Li and Shi, 2014). Both are conventionally driven by a fixed disturbance set and a purely state-based condition. We drive the trigger by perception instead, using the estimator covariance, whose closest precedent is variance-based triggering for state estimation (Trimpe and D'Andrea, 2014), together with the anticipated path curvature, and we certify the resulting inter-trigger behavior.

Building on these observations, we develop COAST, which treats altitude as a coupled perception variable, and make three contributions:

- *The coupling and its regime (Section 3).* We formalize the shared-camera altitude coupling and characterize it in closed form: the estimation-optimal altitude is a curvature-scheduled coverage floor, and we give the exact regime in which adapting altitude helps and the one in which no fixed altitude can be beaten. The contribution is this map of *when* adaptation helps, not a claim that it always does.
- *An event-triggered co-design.* We deploy the policy with a belief-space ground controller and a perception-triggered aerial planner whose ancillary law is shown to be input-to-state stable between triggers and free of Zeno behavior, which makes perception-aware replanning affordable on a constrained companion computer.
- *A reproducible SITL study.* We build a complete ROS 2/Gazebo testbed under one per-step logging protocol and use it to verify each analytical prediction; the code and data are publicly available. [1]

Figure 1 shows the testbed and the single camera in its dual role.

[1] `https://github.com/soododivyam/COAST`



(a) Downward View **(b)** Dual Role

Fig. 1. The single downward-facing camera in its dual role. (a) the aerial view of the marked UGV on its lane. (b) the same frame yields both a navigation waypoint (from the segmented lane) and the EKF pose measurement (from the ArUco marker).

2. SYSTEM MODELING AND PROBLEM FORMULATION

A UAV tracks a UGV in a common world frame $\mathcal{F}_W$. The UAV carries the system's *only* exteroceptive sensor, a single downward-facing monocular camera, while the UGV is otherwise perception-limited, equipped only with wheel odometry. Each camera frame yields two products (Fig. 2): the lane ahead of the vehicle, segmented to produce a navigation waypoint, and an ArUco fiducial marker (Garrido-Jurado et al., 2014) on the UGV, detected to produce a noisy pose measurement. The pose measurement is fused with odometry in an extended Kalman filter (EKF), yielding a belief (μ, Σ), that is, a mean μ and covariance Σ over the UGV state.

Because both products are read from the same image, they share the same altitude, which therefore sets estimation quality and navigation coverage together and in opposite directions. The remainder of this section formalizes the two sides of this coupling through the vehicle motion, resolution, and coverage models, and states the constrained estimation problem they induce, which Section 3 solves.

2.1 Vehicle models

The UGV follows unicycle kinematics with first-order actuator lag. Its state is $\mathbf{x}_g = [x_g, y_g, \theta, v, \omega]^\top$, collecting the planar position (x_g, y_g), the heading θ, and the linear and angular rates (v, ω); the commanded velocities are $\mathbf{u}_g = [v_c, \omega_c]^\top$. The discrete-time model is

$$\mathbf{x}_{g,k+1} = \mathbf{f}_g(\mathbf{x}_{g,k}, \mathbf{u}_{g,k}) + \mathbf{w}_g, \tag{1}$$

where the map $\mathbf{f}_g$ integrates the unicycle kinematics, the actuators track their commands through first-order lags of time constants τ_v, τ_ω, and $\mathbf{w}_g$ is a bounded process noise. The heading and rate states are retained because the marker yields a full planar pose, not merely a position, which the estimator exploits.

The UAV is a quadrotor, but the planning and tracking layers treat it as a velocity-controlled point mass; an inner-loop attitude and thrust controller (in our study, the Gazebo velocity-control plugin) is assumed to realize the commanded velocities. Writing the UAV position as $\mathbf{p} = [p_x, p_y, p_z]^\top$, the altitude as $z := p_z$, and the velocity

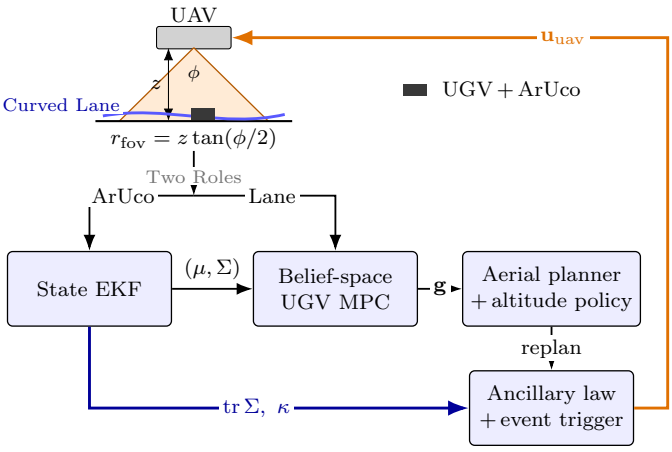


Fig. 2. System overview. The UAV at altitude z views the ArUco-tagged UGV within the footprint $r_{\text{fov}} = z\tan(\phi/2)$, so altitude sets both localization resolution and coverage. The single image yields two products: a lane waypoint and an ArUco pose for the EKF, whose belief (μ, Σ) and the waypoint drive the belief-space UGV MPC; its predicted endpoint $\mathbf{g}$ drives the aerial planner and altitude policy, and the ancillary law and event trigger return the command $\mathbf{u}_{\text{uav}}$ that closes the loop. Blue: the covariance $\text{tr}\,\Sigma$ and curvature κ drive the trigger.

command as $\mathbf{u} = [v_x, v_y, v_z]^\top$, the single-integrator model is

$$\mathbf{p}_{k+1} = \mathbf{p}_k + \Delta t\,\mathbf{u}_k + \mathbf{w}_k, \qquad A_d = I_3, \quad B_d = \Delta t\, I_3, \tag{2}$$

with sampling period Δt, state matrix $A_d = I_3$ (the 3×3 identity), input matrix $B_d = \Delta t\, I_3$, and bounded disturbance $\mathbf{w}_k \in \mathbb{W}$. All three axes are actuated, so the vertical velocity v_z is a genuine decision variable and the altitude is commanded in closed loop. The linear form of (2) makes the ancillary error dynamics exact, which Section 3 uses in the event-triggered controller.

2.2 Resolution model

The accuracy of marker-based localization is set by the camera's ground sampling distance (GSD), the metric size of one pixel on the ground, which grows linearly with altitude. If the image-plane detection error is approximately constant in pixels, the metric position error grows in proportion to the GSD, so the position-measurement covariance grows quadratically with altitude. Let z_0 be a reference altitude, $d = \|\mathbf{p}^{xy} - \mathbf{p}_g\|$ the horizontal UAV–UGV separation (with $\mathbf{p}^{xy}$ the UAV's planar position and $\mathbf{p}_g$ the UGV's), ϕ the camera field of view, and $r_{\text{fov}}(z) = z\tan(\phi/2)$ the resulting footprint radius. We model the measurement covariance as

$$R(z) = R_0 \left(\frac{z}{z_0}\right)^2 \left(1 + \left(\frac{d}{r_{\text{fov}}}\right)^2\right), \tag{3}$$

where R_0 is the baseline covariance at z_0, the factor $(z/z_0)^2$ is the GSD scaling, and the factor $1 + (d/r_{\text{fov}})^2$ mildly inflates the noise as the target approaches the footprint edge (loss of accuracy away from the optical axis). Equation (3) is strictly increasing in z; this monotonicity is the resolution side of the coupling. The scaling is not merely postulated; it is measured directly in Section 4. The same model (3) drives both the EKF and the belief-space planner of Section 3.

2.3 Coverage model

Lane segmentation seeds a waypoint a lookahead distance ahead of the vehicle, and that waypoint must fall inside the footprint or the reference is lost. Writing the required lookahead as $d_{\text{look}}(\kappa)$, which grows with the local path curvature κ (a curving lane leaves the footprint sooner), we summarize navigability by the non-dimensional *coverage factor*

$$\eta(z, \kappa) = \frac{r_{\text{fov}}(z)}{d_{\text{look}}(\kappa)} = \frac{z\tan(\phi/2)}{d_{\text{look}}(\kappa)}, \tag{4}$$

with segmentation feasible for $\eta \geq 1$, equivalently $z \geq z_{\text{fl}}(\kappa) := d_{\text{look}}(\kappa)/\tan(\phi/2)$. This altitude *floor* $z_{\text{fl}}(\kappa)$ is the coverage side of the coupling, and it opposes the resolution side directly: the estimator prefers *low* altitude for resolution (3), whereas (4) demands *sufficient* altitude for coverage, and the floor rises exactly where the path curves.

2.4 Problem formulation

Together the resolution and coverage models pose an altitude-selection problem. Over a traversal of duration T with curvature profile $\kappa(t)$, define the time-averaged estimation cost

$$J[z(\cdot)] = \frac{1}{T}\int_0^T \text{tr}\, P_\infty\big(R(z(t))\big)\, dt, \tag{5}$$

where $P_\infty(R)$ is the steady-state estimation-error covariance under measurement noise R (made precise in Section 3) and $\text{tr}(\cdot)$ is the trace. The *altitude-selection problem* is

$$\min_{z(\cdot)} \ J[z(\cdot)] \quad \text{s.t.} \quad z(t) \geq z_{\text{fl}}(\kappa(t)), \quad z(t) \in [z_{\min}, z_{\max}], \tag{6}$$

that is, minimize the estimation cost subject to coverage feasibility and the actuation limits $[z_{\min}, z_{\max}]$. Alongside (6), the deployed system must (i) keep the UGV within the footprint so that Σ stays bounded, (ii) track the UGV robustly under the bounded disturbance set $\mathbb{W}$, and (iii) limit the frequency of online re-optimization. Section 3 solves (6) in closed form, characterizes when altitude adaptation is worthwhile, and develops the event-triggered controller meeting (i)–(iii).

3. CONTROL DESIGN AND ANALYSIS

This section solves the altitude-selection problem (6) and develops the closed-loop controller: the estimation-optimal altitude and when adapting it beats any fixed choice (Section 3.1), the belief-space UGV controller (Section 3.2), the aerial planner that realizes the altitude policy (Section 3.3), and the event-triggered ancillary controller whose inter-trigger error is provably bounded (Section 3.4).

3.1 Optimal altitude and when adaptation helps

The UGV state is estimated by an extended Kalman filter, whose steady-state behavior we analyze on the linearized model. Let F and H denote the state-transition and measurement Jacobians and Q the process-noise covariance.

Assumption 1. (F, H) is detectable and $(F, Q^{1/2})$ is stabilizable, so that for every measurement-noise covariance $R \succ 0$ the filtering discrete algebraic Riccati equation (DARE)

$$P = FPF^\top + Q - FPH^\top\big(HPH^\top + R\big)^{-1}HPF^\top \quad (7)$$

admits a unique stabilizing solution $P_\infty(R) \succeq 0$.

The estimation cost (5) depends on altitude only through $R(z)$, so its behavior in z is governed by how the steady-state covariance P_∞ depends on R.

Lemma 1. (Monotonicity of the DARE solution). Under Assumption 1, P_∞ is nondecreasing in the measurement noise: if $0 \prec R_1 \preceq R_2$, then $P_\infty(R_1) \preceq P_\infty(R_2)$, and hence $\operatorname{tr} P_\infty(R_1) \le \operatorname{tr} P_\infty(R_2)$.

Proof. In information form the one-step Riccati update has posterior covariance $(P^{-1} + H^\top R^{-1} H)^{-1}$, which is nondecreasing in both P and R, since a larger R subtracts less information; congruence by F and adding Q preserve this order. Iterating from $P_0 = 0$ therefore keeps $R_1 \preceq R_2 \Rightarrow P_k(R_1) \preceq P_k(R_2)$ at every step, and the iterates converge to the stabilizing solution (Anderson and Moore, 1979), so the ordering holds in the limit; this is the comparison property of the discrete-time algebraic Riccati equation (Lancaster et al., 1987; Ran and Vreugdenhil, 1988). The trace inequality follows from $M \succeq 0 \Rightarrow \operatorname{tr} M \ge 0$. □

Since $R(z)$ in (3) is strictly increasing in z, Lemma 1 renders the integrand of (5) monotone in altitude, and the pointwise structure of (6) gives the optimum.

Proposition 1. (Estimation-optimal altitude). Let $g(z) := \operatorname{tr} P_\infty(R(z))$. Under Assumption 1, g is nondecreasing, and the solution of (6) at curvature κ is the coverage floor

$$\begin{aligned} z^\star(\kappa) &= \max\big\{z_{\min},\, z_{\mathrm{fl}}(\kappa)\big\} \\ &= \max\big\{z_{\min},\, d_{\mathrm{look}}(\kappa)/\tan(\phi/2)\big\}. \end{aligned} \quad (8)$$

Proof. Substituting $r_{\mathrm{fov}} = z\tan(\phi/2)$ into (3) cancels the altitude in the off-axis term, giving $R(z) = R_0\,[z^2 + d^2/\tan^2(\phi/2)]/z_0^2$, a fixed positive matrix scaled by a strictly increasing scalar (with d, ϕ exogenous). Hence g is nondecreasing by Lemma 1. Minimizing a nondecreasing g over the feasible interval $[\max\{z_{\min}, z_{\mathrm{fl}}(\kappa)\},\, z_{\max}]$, nonempty since $z_{\mathrm{fl}}(\kappa) \le z_{\max}$, attains the minimum at the left endpoint, which is (8). □

Corollary 1. (Climb into turns). If d_{look} is nondecreasing in κ, then $z^\star(\kappa)$ is nondecreasing in κ: the optimal altitude rises exactly where the path curves.

Proposition 1 gives the optimum at each instant. The operational question is whether adapting altitude along a trajectory beats the best fixed altitude, and when.

Proposition 2. (Adaptive dominance and its regime). Overrier). a traversal of duration T with curvature profile $\kappa(t)$, the curvature-scheduled policy $z^\circ(t) = z^\star(\kappa(t))$ minimizes the cost (5) among all feasible altitude policies. Any feasible constant altitude z_c satisfies $z_c \ge \bar{z} := \max\{z_{\min}, z_{\mathrm{fl}}(\kappa_{\max})\}$, with $\kappa_{\max} = \sup_t \kappa(t)$, so that

$$J[z_c] - J[z^\circ] \;\ge\; \frac{1}{T}\int_0^T \big[g(\bar{z}) - g(z^\circ(t))\big]\,dt \;\ge\; 0. \quad (9)$$

The inequality is strict, so the adaptive policy strictly outperforms every fixed altitude, if and only if the coverage floor binds, $z_{\mathrm{fl}}(\kappa_{\max}) > z_{\min}$, and the curvature is nonconstant, $\kappa(t) < \kappa_{\max}$ on a set of positive measure.

Proof. The integrand $g(z(t))$ is minimized pointwise by the smallest feasible value $z^\circ(t)$, so z° minimizes J. A constant altitude is feasible only if $z_c \ge \sup_t z_{\mathrm{fl}}(\kappa(t))$ and $z_c \ge z_{\min}$, that is $z_c \ge \bar{z}$; monotonicity of g then gives $g(z_c) \ge g(\bar{z}) \ge g(z^\circ(t))$ pointwise, and integration yields (9). The last integral is positive if and only if $z^\circ(t) < \bar{z}$ on a set of positive measure, which requires both $z_{\mathrm{fl}}(\kappa_{\max}) > z_{\min}$ (otherwise $\bar{z} = z_{\min} = z^\circ$ almost everywhere) and $\kappa(t) < \kappa_{\max}$ on positive measure. □

The gap in (9) widens as $z^\circ(t)$ falls further below $\bar{z}$, so adaptation helps most when the curvature-scheduled altitude varies widely. The optimum z° assumes altitude can track the schedule, which the feed-forward policy of Section 3.3 approximates within the UAV's vertical-rate limit.

Corollary 2. (Regime boundary). Adaptation strictly helps if and only if $d_{\mathrm{look}}(\kappa_{\max}) > z_{\min}\tan(\phi/2)$; the separating boundary is $d_{\mathrm{look}}(\kappa_{\max}) = z_{\min}\tan(\phi/2)$.

Proposition 2 answers the adaptive-versus-fixed question in advance and, as Section 4 confirms, predicts the measured outcome: when the required lookahead keeps the floor below $z_{\min}$, no fixed altitude is dominated ($\Delta = 0$); once it exceeds $z_{\min}\tan(\phi/2)$, adaptation strictly helps.

3.2 Belief-space UGV control

The UGV controller augments its planning state with the estimator covariance, $\xi_k = (\mu_k, \Sigma_k)$, and propagates both through the horizon. At each predicted step it forms the no-observation prediction $\Sigma_{\mathrm{pred}} = F\Sigma_k F^\top + Q$ and, when the target is in view, the post-measurement covariance Σ_{obs} from the Joseph-form Kalman update. These are blended by a smooth visibility gate $\gamma = 1/\big(1 + e^{\beta(d - r_{\mathrm{fov}})}\big)$, with steepness $\beta > 0$, which switches from prediction to update as the target enters the footprint:

$$\Sigma_k = (1-\gamma)\,\Sigma_{\mathrm{pred}} + \gamma\,\Sigma_{\mathrm{obs}}. \quad (10)$$

The control minimizes tracking error while penalizing predicted uncertainty and field-of-view (FoV) exit,

$$\begin{aligned} J_g = \sum_{k=0}^{N_g-1} \Big(&\|\mathbf{e}_{g,k}\|^2_{W_e} + \|\mathbf{u}_{g,k}\|^2_{W_u} \\ &+ \lambda\,\operatorname{tr}(\Sigma_k) + w_{\mathrm{fov}}\,c_{\mathrm{fov}}(\mathbf{x}_{g,k})\Big), \end{aligned} \quad (11)$$

over N_g steps, where $\mathbf{e}_{g,k}$ is the waypoint-tracking error weighted by $W_e \succ 0$, $\mathbf{u}_{g,k}$ the input weighted by $W_u \succ 0$, $\lambda > 0$ the covariance-penalty weight, and $c_{\mathrm{fov}} = \big(\max(0,\, d_k - r_{\mathrm{fov}})\big)^2$ a one-sided quadratic hinge (a soft FoV constraint of weight $w_{\mathrm{fov}} > 0$, not an asymptotic bar-

rier). The rollout (10) uses the same measurement model $R(z)$ as the onboard EKF, so the planner propagates exactly the uncertainty the estimator experiences; the terms $\lambda\operatorname{tr}(\Sigma_k)$ and $w_{\mathrm{fov}}c_{\mathrm{fov}}$ are what couple the controller to perception. The covariance rollout makes (11) a nonconvex program, solved by sequential quadratic programming.

3.3 Aerial planner and altitude policy

The UAV computes a nominal trajectory $\bar{\mathbf{p}}$ over N_a steps by trading off goal tracking, an altitude regularizer, and

a visibility term that tolerates lateral offset only when altitude is sufficient:

$$J_a = \sum_k w_{\ell,k}\Big[\|\bar{\mathbf{p}}_k - \mathbf{g}\|^2 + w_z\,(z_k - z^\star)^2 + w_{\text{alt}}\,\frac{\|\bar{\mathbf{p}}_k^{xy} - \mathbf{g}^{xy}\|^2}{z_k^2 + \varepsilon}\Big] + w_u\|\bar{\mathbf{u}}_k\|^2, \tag{12}$$

where $\mathbf{g}$ is the UGV's predicted end-of-horizon position (read from the belief-MPC rollout, one direction of the coupling), $\bar{\mathbf{p}}^{xy}$ and $\mathbf{g}^{xy}$ are the planar components, $w_{\ell,k}$ up-weights later steps, $w_z, w_{\text{alt}}, w_u > 0$ are cost weights, and $\varepsilon > 0$ regularizes the visibility term to avoid the singularity as $z \to 0$. The setpoint $z^\star$ realizes the coverage floor of Proposition 1: the deployed altitude policy

$$z_k = \hat{d}_k/\tan(\phi/2) + \beta_\kappa\,\hat{\kappa}_k + z_{\text{m}} \tag{13}$$

sets altitude from the estimated horizontal UAV-UGV offset $\hat{d}_k$ (the first term keeps the target in the footprint), the anticipated path curvature $\hat{\kappa}_k$ scaled by a feed-forward gain $\beta_\kappa \geq 0$ (which raises altitude ahead of turns, where the required lookahead, and hence the floor, grow), and a fixed margin $z_{\text{m}} > 0$ that keeps the forward lookahead in view on straights. Climbing into turns is thus an emergent consequence of the coverage constraint (Corollary 1), not a hand-coded rule.

3.4 Event-triggered ancillary control

Between re-plans the UAV tracks the nominal trajectory with an ancillary feedback law $\mathbf{u}_k = \bar{\mathbf{u}}_k - K\mathbf{e}_k$, where $\mathbf{e}_k = \mathbf{p}_k - \bar{\mathbf{p}}_k$ is the tracking error and K the feedback gain. Substituting into (2) gives the closed-loop error dynamics

$$\mathbf{e}_{k+1} = (A_d - B_d K)\mathbf{e}_k + \mathbf{w}_k =: A_K\mathbf{e}_k + \mathbf{w}_k, \qquad \mathbf{w}_k \in \mathbb{W}, \tag{14}$$

with $A_K = I_3 - \Delta t\,K$ from (2). The gain K is chosen so that A_K is Schur. A design-time tube-MPC calculation (Mayne et al., 2005; Raković et al., 2005) yields a robust positively invariant set for (14), from which we extract a positive-definite shape matrix P_0 defining the ellipsoid $\mathcal{E}(\alpha) = \{\mathbf{e} : \mathbf{e}^\top P_0\mathbf{e} \leq \alpha^2\}$. Rather than propagate a polyhedral tube online at the ancillary rate, we monitor an adaptive ellipsoid whose radius α_k tightens when the estimate is sharp and expands when perception degrades,

$$\alpha_k = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \times \text{clip}\left(\frac{\operatorname{tr}\Sigma_{xy} - \tau_{\min}}{\tau_{\max} - \tau_{\min}},\, 0,\, 1\right), \tag{15}$$

where $\operatorname{tr}\Sigma_{xy} = \Sigma_{xx} + \Sigma_{yy}$ is the horizontal covariance trace, $[\alpha_{\min}, \alpha_{\max}]$ the admissible radius range, and $[\tau_{\min}, \tau_{\max}]$ the covariance range over which it scales. This surrogate keeps the offline geometry through P_0 while relaxing the recursive-feasibility guarantees of a formal tube MPC in exchange for a lightweight real-time monitor.

The aerial planner re-solves (12) only when one of four perception-driven conditions fires: (i) *tube exit*, $\mathbf{e}_k^\top P_0\mathbf{e}_k > \alpha_k^2$ (critical, no lockout); (ii) *covariance spike*, a sudden rise of $\operatorname{tr}\Sigma_{xy}$ above a threshold, as at marker occlusion; (iii) *curvature anticipation*, the discrete predicted-path curvature $\hat{\kappa} = \max_i |\Delta\theta_i|/\Delta s_i$, with $\Delta\theta_i$ the wrapped heading change over the spacing Δs_i between consecutive predicted waypoints, exceeding a threshold τ_κ, so the UAV climbs before the turn; and (iv) *temporal drift*, an elevated $\operatorname{tr}\Sigma_{xy}$ with time since the last re-plan exceeding $T_{\min}$. Conditions (ii)–(iv) carry a fixed time lockout that guards against chattering; the critical condition (i) does not. Between triggers the ancillary law regulates the error through (14) and the previous nominal sequence is shifted forward.

Two elementary facts make the trigger well posed. First, unrolling the contractive error map (14) and taking induced ∞-norms gives, for $\|A_K\|_\infty < 1$ and $\mathbf{w}_k \in \mathbb{W}$ with $\|\mathbf{w}_k\|_\infty \leq \bar{w}$,

$$\|\mathbf{e}_k\|_\infty \leq \|A_K\|_\infty^k\,\|\mathbf{e}_0\|_\infty + \frac{\bar{w}}{1 - \|A_K\|_\infty}, \tag{16}$$

the input-to-state-stability estimate specialized to a linear map (Jiang and Wang, 2001), so the inter-trigger error stays within the ultimate bound $\bar{w}/(1 - \|A_K\|_\infty)$.

Remark 1. (Boundedness and no Zeno). The bound (16) is a property of the ancillary law: for any $\mathbf{w}_k \in \mathbb{W}$ it holds the error within a fixed ultimate bound (stated for the unsaturated command, as saturation only enlarges that bound). Moreover, the four conditions are evaluated only at the sampling instants $t_k = k\Delta t$, so successive firings are separated by at least Δt; the minimum inter-event time is thus $\Delta t > 0$ and Zeno behavior is excluded by construction (Heemels et al., 2013). Ellipsoid membership (15) is therefore a well-posed firing condition, and re-plans are driven by genuine perception and curvature excursions rather than by chattering.

4. SIMULATION STUDY

The aim of this section is not to tune the controller but to test the analysis of Section 3: each experiment isolates one result and checks it against measured closed-loop data. We proceed in the order of the theory, verifying the resolution model behind Assumption 1 (Section 4.2), the estimation-cost monotonicity that Lemma 1 and Proposition 1 rest on (Section 4.3), the regime prediction of Proposition 2 for altitude adaptation (Section 4.4), and the compute saving of the event-triggered co-design (Section 4.5).

4.1 Testbed and protocol

The system runs in Gazebo with ROS 2 (Macenski et al., 2022). A TurtleBot3-class UGV carrying an ArUco marker is tracked by an X500-class UAV with a downward camera; an OpenCV lane and marker pipeline, the five-state EKF, the belief-space UGV MPC of (11), and the UAV planner and ancillary pair of (12) and (14) run as separate nodes. A single logger, identical across every configuration, records per-step pose, ground truth, $\operatorname{tr}\Sigma$, footprint distance, ArUco visibility, replan events, and solve times, so that all configurations are compared on the same measurements. Experiments use a standard set of tracks of increasing curvature, denoted Low-κ, Mid-κ, High-κ, and a variable-curvature Var-κ track whose curvature alternates along its length. All numeric parameter values are collected in Table 1, including the visibility regularizer $\varepsilon = 0.1$ of (12).

4.2 Resolution model and the estimator premise

Assumption 1 and the resolution model (3) rest on one empirical claim, that measurement noise grows with altitude

Table 1. System Parameters and Controller Hyperparameters

Subsystem	Parameter	Value
Camera / EKF	Camera FoV (ϕ)	1.047 rad
	Baseline altitude (z_0)	3.0 m
	Baseline covariance (R_0)	2.5×10^{-3} m^2
	EKF update rate	20 Hz
UGV MPC	Horizon N_g, step Δt	5, 0.2 s
	Weights $W_e/W_u/w_{\text{fov}}$	10 / 10 / 5
	Covariance weight λ	1.0
	Bounds $v_{\max}$, $\omega_{\max}$	0.22 m/s, 2.0 rad/s
UAV planner	Horizon N_a, step Δt	10, 0.2 s
	Velocity bounds (v_x, v_y, v_z)	$\pm 1.5, \pm 1.5, \pm 1.0$ m/s
	Altitude weight w_{alt}	50
	Policy z_{m}, β_κ	0.4 m, 5.0
	Regularizer ε	0.1
Tube & triggers	Shape matrix P_0	$0.186\, I_3$
	Tube radius $[\alpha_{\min}, \alpha_{\max}]$	$[0.5, 2.0]$
	Cov. triggers (spike, drift)	0.12, 0.07
	Timing ($T_{\min}$, lockout)	3.0 s, 1.0 s
	Altitude band $[z_{\min}, z_{\max}]$	$[2.5, 7.0]$ m

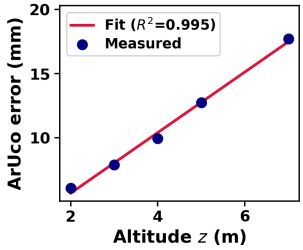


Fig. 3. Resolution-model validation (static altitude sweep, marker on axis). The mean ArUco position error grows linearly with altitude (17) ($R^2 = 0.995$), confirming the $R \propto z^2$ law of (3). One altitude with an intermittent detection dropout is excluded from the fit.

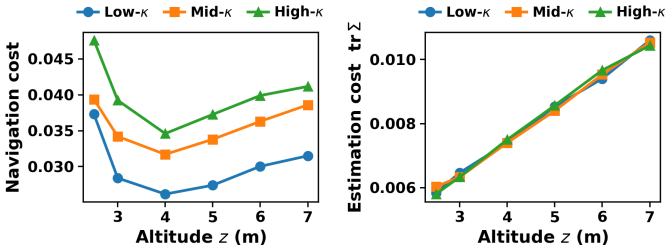
as $R \propto z^2$. We test it directly. With the UAV held static at a sequence of altitudes and the marker near the optical axis ($d \approx 0$, so the off-axis factor of (3) is inactive), we log the raw ArUco position error over roughly 150 detections per altitude. The mean error grows linearly with altitude,

$$\hat{e}(z) = 0.94 + 2.36\, z \ \ [\text{mm}], \qquad r = 0.997, \quad R^2 = 0.995, \tag{17}$$

over $z \in [2, 7]$ m (Fig. 3). Because the pixel-level detection error is altitude-invariant while the ground sampling distance scales linearly with z, a linear growth of the metric error is exactly a variance that scales as z^2, which is the $R \propto z^2$ law of (3). The fit in (17) excludes a single altitude at which the marker was intermittently lost, a detection dropout that inflated the sample variance by two orders of magnitude. This establishes the resolution side of the coupling as a measured property of the sensor rather than a modeling assumption, and with it the well-posedness premise on $R \succ 0$ that Assumption 1 requires.

4.3 Estimation cost and navigation cost versus altitude

Lemma 1 predicts that the steady-state estimation cost $g(z) = \operatorname{tr} P_\infty(R(z))$ is nondecreasing in altitude, and Proposition 1 builds the optimal-altitude result on top of



(a) Navigation Cost **(b)** Estimation Cost

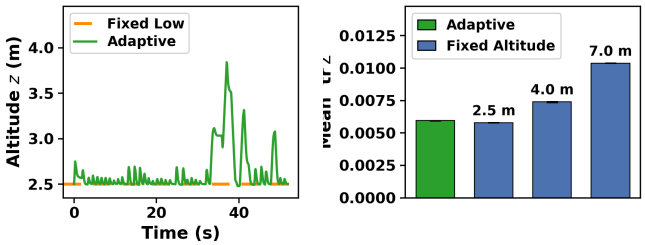
Fig. 4. Coverage versus resolution across altitude. **(a)** navigation (steering) cost is smallest at intermediate altitude and rises at both extremes, growing with curvature κ. **(b)** estimation cost $\operatorname{tr}\Sigma$ grows monotonically with altitude (Lemma 1) and is track-independent. The optimum thus flies as low as coverage allows (Proposition 1).

it. We verify the premise in closed loop by sweeping the commanded altitude over $[2.5, 7]$ m on the Low-κ, Mid-κ, and High-κ tracks with the GSD noise model active. The measured steady-state covariance trace rises monotonically with altitude, from $\operatorname{tr}\Sigma = 0.0059$ at $z = 2.5$ m to 0.0105 at $z = 7$ m (Fig. 4(b)), and at each altitude it is nearly independent of the track: for example $\{0.0058, 0.0060, 0.0058\}$ across the three tracks at $z = 2.5$ m and $\{0.0106, 0.0105, 0.0104\}$ at $z = 7$ m. This track-independence is exactly what the shared-camera model predicts, since the estimation cost enters through $R(z)$ alone and not through path geometry. The opposing consideration is navigation. The UGV steering effort needed to keep the lane segmented is smallest at intermediate altitude and rises at both extremes (Fig. 4(a)): steeply as altitude drops toward the coverage floor, where the shrinking footprint begins to drop the lookahead, and mildly at high altitude, where the coarser sampling adds waypoint noise. It also rises with curvature at every altitude. Estimation cost, by contrast, decreases monotonically as altitude falls. Together the two express the coverage-resolution tradeoff of (3) and (4): estimation favors low altitude while coverage forbids flying below the floor, which is why the estimation-optimal altitude of Proposition 1 flies as low as segmentation allows and climbs where the floor rises.

4.4 Altitude adaptation and its regime

Proposition 2 gives a testable prediction: the curvature-scheduled policy strictly beats every fixed altitude if and only if the coverage floor binds, $z_{\text{fl}}(\kappa_{\max}) > z_{\min}$; otherwise it ties the lowest feasible fixed altitude, with gap $\Delta = 0$. Our deployed configuration sits on the non-binding side of the Corollary 2 boundary. The required lookahead is $d_{\text{look}} \approx 0.5$ m, so the floor is $z_{\text{fl}} \approx d_{\text{look}}/\tan(\phi/2) \approx 0.87$ m, well below the actuation limit $z_{\min} = 2.5$ m. The proposition therefore predicts $\Delta = 0$ here, which is what the experiment should show.

We run the policy against three fixed altitudes (2.5, 4.0, and 7.0 m) on the Var-κ track, three repetitions each. The results match the prediction (Fig. 5):



(a) Altitude vs. Time **(b)** Mean Estimation Cost

Fig. 5. Altitude adaptation in the non-binding regime (Var-κ track, three repetitions). **(a)** the adaptive altitude holds near the floor on straights and climbs into turns (Corollary 1), spanning $[2.5, 3.9]$ m. **(b)** mean estimation cost $\mathrm{tr}\,\Sigma$: the adaptive policy matches the lowest fixed altitude ($\Delta \approx 0$, as Proposition 2 predicts for $z_{\mathrm{fl}} < z_{\min}$), while higher fixed altitudes cost up to 79% more.

- The adaptive policy attains mean $\mathrm{tr}\,\Sigma = 0.0060$, only 2.8% above the lowest fixed altitude (0.0058 at $z = 2.5$ m), so $\Delta \approx 0$;
- The fixed-altitude cost rises monotonically with height, $0.0058 \to 0.0074 \to 0.0104$ from 2.5 to 7 m, a 79% estimation penalty at the top, exactly the $g(z)$ increase of Lemma 1;
- The adaptive altitude is not constant: it holds near the floor on straights and climbs to 3.9 m through the sharpest turns, the "climb into turns" behavior of Corollary 1 emerging from the coverage term rather than from a scripted maneuver;
- Every configuration keeps the marker in view (ArUco visibility $\geq$ 99%, no off-corridor waypoints), consistent with a regime in which even the lowest altitude has adequate coverage.

The small gap is not a failure but the proposition's prediction for this regime: with the floor below $z_{\min}$, no fixed altitude is dominated, so adaptation essentially matches the best fixed choice, the residual reflecting the curvature-driven climbs the policy still commands where coverage does not require them, while still avoiding the estimation penalty any higher altitude pays. The complementary binding regime, $z_{\mathrm{fl}}(\kappa_{\max}) > z_{\min}$, where the gain turns strictly positive, is entered only by a lookahead that scales with speed or curvature; we characterize it in closed form and leave its closed-loop demonstration to future work.

4.5 Event-triggered co-design

Remark 1 shows that the perception-driven trigger fires without Zeno behavior and keeps the inter-trigger error bounded, making it safe to replace continuous aerial replanning by event-triggered replanning. The payoff is compute. Because the belief-space UGV controller runs continuously in every configuration, the lever is the UAV nominal planner (12); we compare continuous replanning (C1) against the event-triggered scheme (C2) at a fixed altitude ($z = 6$ m), so that the compute effect is isolated from the altitude policy. Over three repetitions (Table 2), C2 reduces aerial solves by 24.9× ($797 \pm 16 \to 32 \pm 1$ per run) and aerial CPU by 12.3× ($3123 \pm 56 \to 253 \pm 6$ ms per run), at matched estimation quality: the steady-state covariance trace is 0.0095 for both schemes and ArUco visibility is $\approx 99.9\%$ for both. Each individual C2 solve costs about twice as much (7.9 ± 0.4 vs. 3.9 ± 0.1 ms), because it re-optimizes from a shifted nominal trajectory, but the near 25-fold reduction in the number of solves dominates by an order of magnitude. The firings are dominated by curvature-anticipation events with occasional plan-expiry events, and the re-plan rate stays bounded, consistent with Remark 1. This realizes the deployability requirement (iii) of Section 2.4, perception-aware aerial replanning at the cost of an occasional solve rather than a continuous one.

Table 2. Aerial-planner co-design (fixed altitude $z = 6$ m, mean $\pm$ std over three repetitions). C1 continuous vs. C2 event-triggered; the belief-space UGV controller runs continuously in both. Steady-state window $t \geq 8$ s.

Metric	C1 (cont.)	C2 (event)
Aerial solves (#/run)	797 ± 16	32 ± 1
Mean solve (ms)	3.9 ± 0.1	7.9 ± 0.4
Aerial CPU (ms/run)	3123 ± 56	253 ± 6
Steady-state $\mathrm{tr}\,\Sigma$	0.0095	0.0095
ArUco vis. (%)	≈ 99.9	≈ 99.9
UAV–UGV offset (m)	0.128 ± 0.002	0.060 ± 0.005

5. CONCLUSION AND FUTURE WORK

We studied aerial-ground cooperative tracking in which a single downward UAV camera is the ground vehicle's only exteroceptive sensor, serving at once as its navigation reference and as the input to its state estimator. The central observation is that this shared-sensor design couples navigation and estimation through one scalar, the UAV altitude, which sets footprint coverage and localization resolution in opposite directions. Treating altitude as an active perception variable, we characterized in closed form the estimation-optimal altitude as the curvature-scheduled coverage floor (Proposition 1) and the exact regime in which adapting it is worthwhile (Proposition 2): adaptation strictly dominates every fixed altitude if and only if the coverage floor binds and the curvature varies. A perception-triggered co-design deploys this policy with a bounded, Zeno-free inter-trigger error (Remark 1). The SITL study confirms each prediction: the sensor follows $R \propto z^2$, the estimation cost is monotone and track-independent in altitude, the curvature-scheduled policy matches the best feasible fixed altitude in our non-binding configuration while avoiding the estimation penalty of flying higher, and the event trigger reduces aerial-replanning compute by more than an order of magnitude at unchanged estimation quality (Table 2). The characterization of when altitude adaptation helps, not any single altitude value, is the contribution.

Several aspects of the present study bound its scope and set the agenda for future work.

- *Binding regime.* Our testbed sits on the non-binding side of the Corollary 2 boundary, where the required lookahead keeps the coverage floor below $z_{\min}$, so the strict-dominance benefit ($\Delta > 0$) is established analytically but not yet demonstrated in closed loop. A lookahead that scales with speed or curvature raises the floor above $z_{\min}$, and exhibiting the predicted $\Delta > 0$ under such a law is the natural next experiment.

- *Higher fidelity.* The study is software-in-the-loop with a velocity-controlled UAV abstraction, so an inner-loop attitude and thrust controller is assumed to realize the commanded velocities; replacing this abstraction with a full multicopter model and moving to hardware is the next step.
- *Stronger guarantees.* The controller monitors an adaptive ellipsoidal surrogate rather than propagating a full tube online, and the bound of Remark 1 is stated for the unsaturated command; strengthening the surrogate toward a formal recursive-feasibility argument, and benchmarking the perception-driven trigger against plain error-triggered and self-triggered baselines, would tighten the analysis.
- *Harder perception.* Finally, we plan to evaluate the single-camera architecture under intermittent observations, latency, and occlusions.

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